

IBM i Incident Insights & Response Assistant

PROJECT REQUIREMENTS, SETUP, AND CONFIGURATION GUIDE

1. Credentials and Configuration

1.1 Important Security Notice

Security Notice: This project requires external credentials for Google Gemini AI and Gmail API functionality. For security reasons, the original developer's API keys, OAuth client credentials, authorization tokens, and passwords are not included in the submitted project. Any person running this project must configure their own credentials.

The following credentials are required:

- Google Gemini API Key
- Google OAuth 2.0 Client Credentials for Gmail functionality
- Gmail account authorization

1.2 Gemini API Configuration

The AI Insights feature uses the Google Gemini API. The application expects the Gemini API key to be stored in the following location:

- `.streamlit/secrets.toml`

The file should contain:

```
GEMINI_API_KEY = "YOUR_GEMINI_API_KEY_HERE"
```

- Replace `YOUR_GEMINI_API_KEY_HERE` with your own valid Gemini API key.
- The configuration variable name must remain exactly `GEMINI_API_KEY` because the application reads the API key using the Streamlit secrets configuration.
- The Gemini API key must be kept private and must not be shared publicly or committed to a public GitHub repository.

1.3 Gmail OAuth Configuration

The application includes an optional Gmail integration feature that allows generated AI operational insights to be sent by email.

The Gmail functionality uses:

- Google OAuth 2.0
- Gmail API
- OAuth Desktop Client credentials

The user running the project must create and configure their own Google OAuth credentials. The OAuth credentials file must be placed at:

- `credentials/credentials.json`

The original OAuth credentials used during project development are intentionally not included in the submitted project. After the user completes the Google authorization process for the first time, the application automatically creates an authorization token at:

- `credentials/token.json`

The `token.json` file is specific to the authenticated user and must also remain private.

1.4 Gmail OAuth Setup Summary

1. Create a Google Cloud project.
2. Enable the Gmail API.
3. Configure the OAuth consent screen.
4. Create an OAuth Desktop Client.
5. Download the OAuth client credentials JSON file.
6. Rename or place the file as: `credentials/credentials.json`.
7. Run the application.
8. When the Gmail feature is used for the first time, complete the Google OAuth authorization process in the browser.
9. After successful authorization, the application creates `credentials/token.json` automatically.

2. Software and Project Requirements

2.1 Purpose

Before running the IBM i Incident Insights & Response Assistant, the user must have the required software, Python environment, project files, internet connectivity, and Google services configuration. The application is a Python-based Streamlit application that performs incident data analysis and provides AI-assisted operational insights.

2.2 Required Software

Python must be installed on the computer. The user should ensure that Python and pip are available from the Command Prompt or terminal. The following commands can be used to verify the installation:

```
python --version  
pip --version
```

If Python is not available using the `python` command, the user may need to use the appropriate Python command configured on their operating system.

2.3 Required Python Libraries

The project includes a `requirements.txt` file containing the required Python packages:

- `streamlit`
- `pandas`
- `altair`
- `openpyxl`
- `google-genai`
- `google-api-python-client`
- `google-auth-httpplib2`
- `google-auth-oauthlib`

Library	Purpose
Streamlit	Application user interface and web application framework
Pandas	Incident data processing and analysis
Altair	Data visualization
OpenPyXL	Reading Excel files
Google Gen AI	Google Gemini AI integration
Google API Python Client	Gmail API interaction
Google Auth HTTP Library	Google authentication support
Google Auth OAuth Library	Google OAuth 2.0 authentication

2.4 Install Project Dependencies

After extracting the project ZIP file and opening a terminal or Command Prompt inside the project folder, install the required dependencies using:

```
pip install -r requirements.txt
```

This command installs the Python libraries required by the application.

2.5 Internet Connection

An active internet connection is required for the following functionality:

- Google Gemini AI Insights generation
- Gmail API integration
- Google OAuth authentication

The dashboard and local data analysis depend on the local project files, but AI and Gmail-related functionality require access to Google's online services.

2.6 Google Account

A Google account is required if the user wants to use the complete functionality of the project. The Google account may be used for:

- Creating or accessing a Gemini API key
- Creating a Google Cloud project
- Configuring Gmail API access
- Completing Gmail OAuth authorization

2.7 Required Project Configuration

Before running the complete application, the following configuration must be completed:

- **Required for AI Insights:** Gemini API Key configured in `.streamlit/secrets.toml`
- **Required for Gmail Functionality:** Google OAuth Credentials placed at `credentials/credentials.json`

Detailed instructions for configuring these credentials are provided in the Credentials and Configuration section of this document.

2.8 Browser

A modern web browser is required to access the Streamlit application after it starts. The browser is also used during the Gmail OAuth authorization process.

Summary of Requirements

- ✓ Python installed
- ✓ pip available
- ✓ Project files extracted
- ✓ Python dependencies installed
- ✓ Required incident data file available
- ✓ Gemini API key configured
- ✓ Internet connection available
- ✓ Google account available
- ✓ Gmail OAuth configured (if Gmail functionality is required)
- ✓ Modern web browser available

3. Incident Data Requirements

3.1 Overview

The IBM i Incident Insight & Response Assistant requires an Excel file containing structured incident data. This dataset is the primary data source used by the application to:

- Load and display incident records
- Validate the dataset structure
- Calculate key performance indicators
- Identify operational patterns
- Generate dashboard visualizations
- Produce structured operational findings
- Provide validated findings to Google Gemini for AI-assisted interpretation

The application prototype uses synthetic IBM i operational incident data.

3.2 Required Data File

The application expects the incident data file at the following location:

- `data/IBM_i_Incident_Data.xlsx`

```
Project Files/  
|  
├─ data/  
│   └─ IBM_i_Incident_Data.xlsx  
└─ app.py
```

The filename and location should remain unchanged unless the `DATA_FILE` configuration in `app.py` is updated accordingly.

3.3 Required File Format

The incident dataset must be provided as an Excel file in `.xlsx` format. The application reads the Excel file using the Pandas Excel-reading functionality. Therefore, the required Python dependency `openpyxl` must be installed as part of the project dependencies.

3.4 Required Dataset Columns

Column Name	Description
<code>incident_id</code>	Unique identifier for each incident
<code>timestamp</code>	Date and time associated with the incident
<code>system_name</code>	IBM i system or environment associated with the incident
<code>subsystem</code>	IBM i subsystem related to the incident
<code>alert_category</code>	Category or type of alert
<code>message_id</code>	IBM i message or alert identifier
<code>severity</code>	Severity level of the incident
<code>description</code>	Description of the incident
<code>resolution_status</code>	Current resolution status of the incident
<code>resolution_duration_minutes</code>	Time taken to resolve the incident, measured in minutes
<code>business_impact</code>	Business impact associated with the incident
<code>affected_users</code>	Number of users affected
<code>assigned_team</code>	Support team assigned to the incident

These column names must be present in the Excel file for the application's dataset validation and analysis to work correctly.

3.5 Data Type Expectations

The application performs conversion of selected columns after loading the dataset:

- **Timestamp:** The `timestamp` column is converted to a date/time format. Values should contain valid date and time information.
- **Numeric Fields:** The `resolution_duration_minutes` and `affected_users` columns are converted to numeric values. These columns should contain numeric values where possible. If invalid values are present, they may be converted to missing values during processing.

3.6 Dataset Validation

After loading the incident dataset, the application performs validation checks:

- **Required Columns:** The application checks whether all required columns are present.
- **Missing Values:** The application checks for missing values across the dataset.
- **Duplicate Incident IDs:** The application checks for duplicate values in `incident_id`.

If required columns are missing, or if data quality issues are identified, the application displays the validation results to the user.

3.7 Data Quality Recommendations

For best results, the incident dataset should:

- Contain all required columns.
- Use the exact column names listed above.
- Contain valid date/time values in the `timestamp` column.
- Contain numeric values in `resolution_duration_minutes`.
- Contain numeric values in `affected_users`.
- Use unique values for `incident_id`.
- Avoid unnecessary missing values.
- Maintain consistent values for categories such as severity and resolution status.

Poor data quality may affect dashboard calculations, operational findings, and AI-assisted insights.

3.8 Using a Different Incident Dataset

A different incident dataset can be used, provided that:

1. The file is in Excel `.xlsx` format.
2. The required columns are present.
3. The column names match the names expected by the application.
4. The file is placed at: `data/IBM_i_Incident_Data.xlsx`.

Alternatively, the dataset filename or location can be changed by updating the following configuration in `app.py`:

```
DATA_FILE = "data/IBM_i_Incident_Data.xlsx"
```

If the dataset structure is significantly different, changes to the application code may also be required.

4. Project Folder Structure and File Purpose

4.1 Overview

The project should maintain the following folder structure to ensure that the application can locate the required data files, prompt files, styles, and configuration files correctly.

```
Project Files/
|
|— .streamlit/
|   └─ secrets.toml
|
|— credentials/
|   └─ README.txt
|
|— data/
|   └─ IBM_i_Incident_Data.xlsx
|
|— prompts/
|   └─ ai_insights_prompt.txt
|
|— Styles/
|   └─ app_styles.css
|
|— .gitignore
|— app.py
|— gmail_service.py
|— README.md
└─ requirements.txt
```

Note: After Gmail OAuth configuration and authorization, additional files may be created inside the credentials folder, including `credentials.json` and `token.json`.

4.2 Root-Level Files

- `app.py` : Main application file. It handles loading the incident Excel data, validating the dataset, processing/analyzing incident data, calculating dashboard metrics, displaying charts/visualizations, generating structured operational findings, calling Google Gemini for AI insights, displaying AI-generated insights, and providing the user interface. This is the main file executed by Streamlit.
- `gmail_service.py` : Contains the Gmail and Google OAuth logic. Manages Google OAuth authentication, loading OAuth credentials, creating/refreshing authorization tokens, connecting to the Gmail API, and sending generated insights via email. Separating this keeps Gmail integration modular and easier to maintain.
- `requirements.txt` : Lists the Python libraries required to run the project via `pip install -r requirements.txt`.
- `README.md` : Provides a general project overview, including project purpose, key features, technology stack, project structure, installation instructions, configuration requirements, security considerations, limitations, and future enhancements.
- `.gitignore` : Prevents sensitive or unnecessary files (e.g., Gemini API keys, Gmail OAuth credentials, authorization tokens, private configuration files) from being committed to Git/GitHub.

4.3 Specialized Folders

- `.streamlit/secrets.toml` : Stores Streamlit secrets, specifically the Gemini API key placeholder (`GEMINI_API_KEY = "YOUR_GEMINI_API_KEY_HERE"`).
- `credentials/` : Reserved for Gmail OAuth credentials (`credentials.json`) and generated session tokens (`token.json`). Both contain sensitive information and must remain private.
- `data/` : Stores `IBM_i_Incident_Data.xlsx`, which provides records for dashboard metrics, operational trends, and AI analysis.
- `prompts/` : Contains `ai_insights_prompt.txt`, keeping prompt template logic clean and modular.
- `Styles/` : Stores `app_styles.css` for application typography, colors, and layout customization.

4.4 Summary of File Responsibilities

File / Folder	Primary Purpose
app.py	Main Streamlit application
gmail_service.py	Gmail API and OAuth functionality
requirements.txt	Required Python dependencies
README.md	General project documentation
.gitignore	Prevents sensitive/unnecessary files from being committed
.streamlit/secrets.toml	Gemini API configuration
credentials/	Gmail OAuth credentials and token files
data/	Incident Excel dataset
prompts/	AI prompt template
styles/	Application CSS styling

5. Project Development Approach and Application Workflow

5.1 Development Approach

The IBM i Incident Insights & Response Assistant was developed using a layered approach that combines traditional data analysis with Generative AI. Structured data analysis is performed before AI-assisted interpretation is introduced.

Data Analysis → Structured Findings → AI-Assisted Interpretation → Human Review

This approach separates factual data processing from AI-generated interpretation.

5.2 Overall Application Workflow



5.3 Application Stages

- **Stage 1 — Incident Data Source:** Primary Excel dataset containing timestamps, alerts, message IDs, severity, impacted users, and assigned teams.
- **Stage 2 — Data Loading and Validation:** Verification of dataset columns, duplicate IDs, missing values, and formatting.
- **Stage 3 — Python and Pandas Data Processing:** Data conversion, type parsing, and preparation for analysis.
- **Stage 4 — Deterministic Operational Analysis:** Reliable, deterministic metric calculation (counts, averages, affected users, system breakdown).
- **Stage 5 — Structured Operational Findings:** Packaging factual metrics into summarized findings rather than sending large, raw tables to the LLM.
- **Stage 6 — Google Gemini AI Interpretation:** Generating executive summaries, identified risks, and recommended actions using structured findings.
- **Stage 7 — Human Review and Validation:** Human-in-the-loop review to ensure operational decisions reflect real organizational context.
- **Stage 8 — Download and Gmail Sharing:** On-demand export and modular email distribution.

6. Installation and Application Startup Instructions

6.1 Extract the Project Files

Extract the submitted project ZIP file to a suitable directory:

```
C:\Projects\IBM_i_Incident_Insights_Assistant\
```

6.2 Complete Required Configuration

1. Configure the Gemini API key in `.streamlit/secrets.toml`.
2. Configure Gmail OAuth credentials in `credentials/credentials.json` if Gmail functionality is required.
3. Ensure the incident Excel file is placed at `data/IBM_i_Incident_Data.xlsx`.

6.3 Command Line Setup & Execution

Navigate to the project root directory and install dependencies:

```
cd C:\Projects\IBM_i_Incident_Insights_Assistant
pip install -r requirements.txt
```

Start the Streamlit application:

```
streamlit run app.py
```

Open the URL displayed in the terminal (typically `http://localhost:8501`) in your browser. To stop the server, press `Ctrl + C` in the terminal.

7. Application Features and User Guide

- **Main Incident Dashboard:** High-level metrics, charts, severity patterns, and subsystem trends.
- **Key Operational Findings:** Factual patterns derived through deterministic analysis.
- **Incident Data Explorer:** Row-level record viewer for underlying operational transparency.
- **AI-Assisted Insights:** Gemini-powered advisory summaries, risks, and recommended priority actions.
- **Download and Sharing:** Insight export and optional Gmail integration.

8. Project Limitations

- Uses an Excel file as the data source rather than a direct connection to a live IBM i database.
- AI Insights require active internet connectivity and Gemini API service availability.
- Analytical results directly depend on the accuracy and completeness of the input data.
- AI-generated insights are advisory and require human review.
- Gmail features require independent Google Cloud OAuth client configuration.

9. Security Considerations

- Never share, commit, or publicly expose Gemini API keys.
- Gmail OAuth credentials and authorization tokens must remain strictly private.
- Each user must configure their own independent API keys and credentials.
- Verify that `.gitignore` is properly maintained to avoid committing sensitive tokens or data to public repositories.